First Order System

Pendulum

$\frac{dx_1}{dt} = x_2$

$\frac{dx_2}{dt} = -\frac{g}{L} sin(x_1)$

What it Models?

Pendulum

The pendulum equation is a way to model a real world pendulum consisting of a weight and massless cord to create a pivot, such as in a simple gravity pendulum. These equations can be used to find the length of cord, gravitational field, and Θ or the angle the pendulum lies at at a given time. x_1 and x_2 each respectively represent Angular Position(1) and Angular Velocity(2).

Parameters

Pendulum

Our parameters for the Pendulum equation are our gravitational field (g) and the length of our chord (L).